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Course Name: Object-Oriented Programming (OOP)

Course Code: CIS-121

Degree Program: BS CIS (23-27)

Department: Department of Computer and Information Sciences (DCIS)

Semester: 2nd Semester

Semester Project:

Brick Breaker Game in Java

Members:

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Initial Report

Description:

My project is a classic **arcade-style game** called Brick Breaker. It was originally developed by Atari Inc. in 1976. **In this game**, players control a paddle at the bottom of the screen to bounce a ball and break bricks arranged at the top of the screen. The goal is to clear all the bricks using the ball without letting it fall off the bottom of the screen.

This project is a game developed using Java programming language, JFrame, JPanel, 2D Graphics and java Graphical-User Interface (GUI).

Features and Functions:

1. **Paddle Control:**

Players can move the paddle horizontally using the left and right arrow keys to prevent the ball from falling off the screen.

2. **Ball-Bouncing Physics:**

The ball moves around the screen, bouncing off the walls, paddle, and bricks, following the laws of physics like changing directions, collision detection and so on.

3. **Brick Layout:**

Bricks are arranged in various patterns at the top of the screen, and players aim to break them all by hitting them with the ball.

4. **Scoring System:**

Players earn points for each brick they break. The score is displayed on the screen.

5. **Game Over Conditions:**

The game ends when the player clears all the bricks or when the ball falls off the bottom of the screen. In either case, the game displays a game over message and allows the player to restart the game.

6. **Single-Player Mode:**

The game supports single-player mode, where one player controls the paddle to break bricks. It means only one player can play at a time.

Future Enhancements:

In future iterations of the project, I plan to add additional features such as:

- Power-ups and special abilities

- Multiple levels with increasing difficulty

- High score tracking and leaderboard

- Sound effects and background music

It’s all will be done using some kind of game engines like unity platform.

Programming Platform that I Will Use:

To run the Brick Breaker game that I have developed, I will use a **Java Development Kit (JDK)**. I can compile and run the Java source code using a Java Integrated Development Environment **(IDE)** such as **Eclipse**. Eclipse provides a user-friendly interface for writing, compiling, and running Java programs, making it easy to work with my project.

Conclusion:

In the conclusion, my Brick Breaker game offers classic arcade entertainment with amazing gameplay mechanics and intuitive controls. It provides players with a challenging yet enjoyable experience. **In this game**, players control a paddle at the bottom of the screen to bounce a ball and break bricks arranged at the top of the screen. The goal is to clear all the bricks using the ball without letting it fall off the bottom of the screen.